Travelly

One Trip One Place Zero Stress

CS253 Project by Team 17 **(10 Bit)**

github.com/CS253

Frontend
Backend
Documents





The Friction in Group Travel

Financial Friction

Splitting expenses manually creates a web of confusion

Coordination Chaos

Itineraries, documents and updates scattered across apps

No Unified Solution

Bills, planning, and chat - fragmented across multiple tools

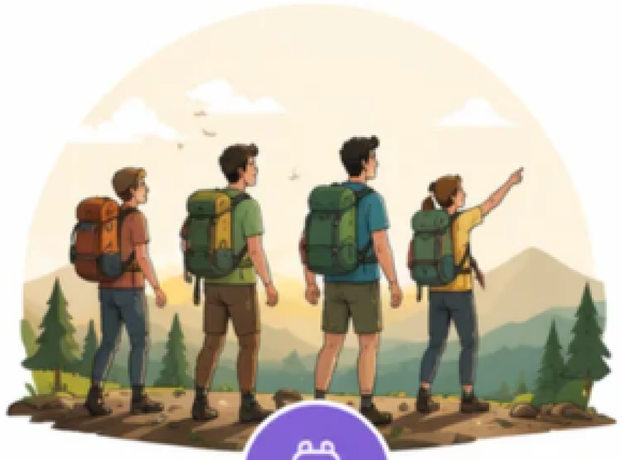
Who is Travelly For?

Built for every kind of group trip, big or small



Students & Friend Groups

Weekend getaways on tight budgets... every dollar tracked



Organizers & Backpackers

Group hikes and events — logistics, funds, and debts managed



Families & Relatives

Large reunions with expenses split across households



Requirements: Smart Expense Tracking

Expense Logging & Bill Splitting

Equal & unequal splits - percentages, exact amounts, shares

Debt Simplification Algorithm

Optimal settlement paths - e.g. 6 debts reduced to 2

Shared Media Gallery

Centralized trip photos & videos for all members



Requirements: Trip Management & Collaboration

Instant Trip Onboarding

Add with Phone Numbers and Deep-link invites

Dynamic Group Management

Multiple concurrent trips, scoped expenses & media

One-Stop Ecosystem

Finance, coordination & media, one cross-platform app



Implementation

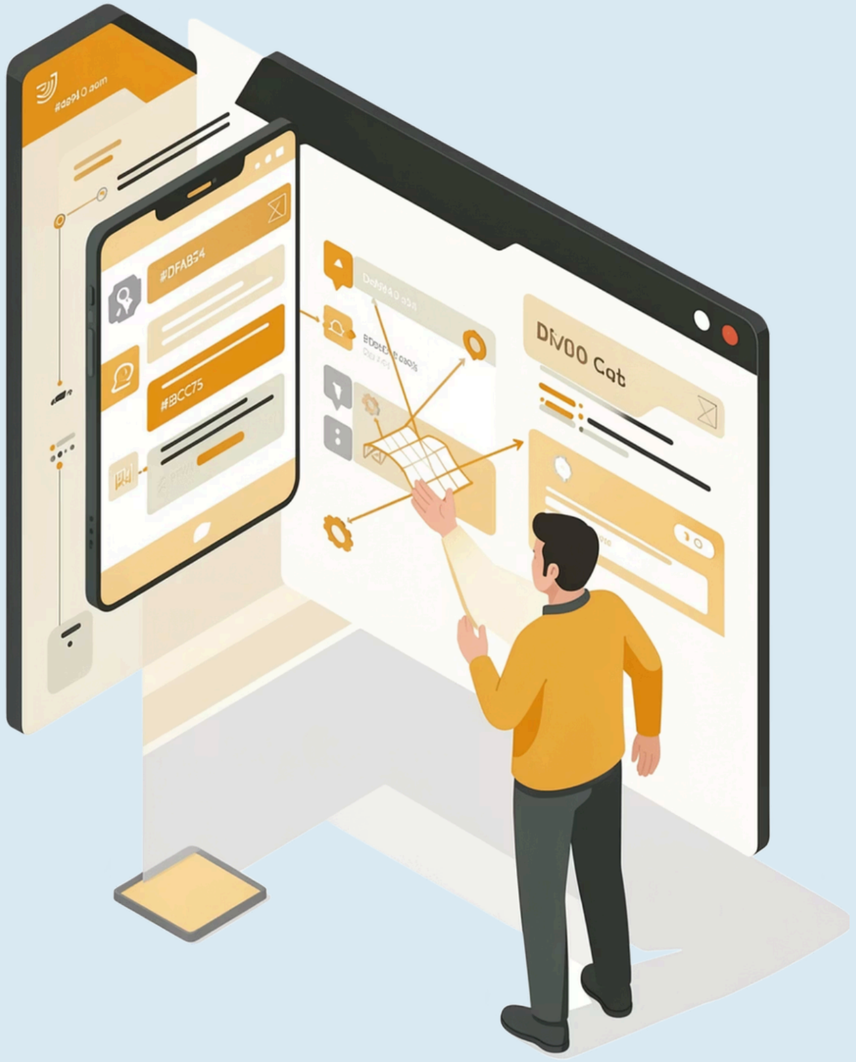
Design: We used **Figma** to design the user interface, create wireframes, and plan the overall user experience before development.

Frontend: The frontend was developed using **Flutter**, which helped us build a responsive and smooth cross-platform application interface.

Backend: The backend was implemented using **Node.js** and **Express.js**, which were used to handle server-side logic, APIs, and application flow efficiently.

Database: We used **PostgreSQL** as our primary database, leveraging the Neon interface for collaboration, management, and scalability.

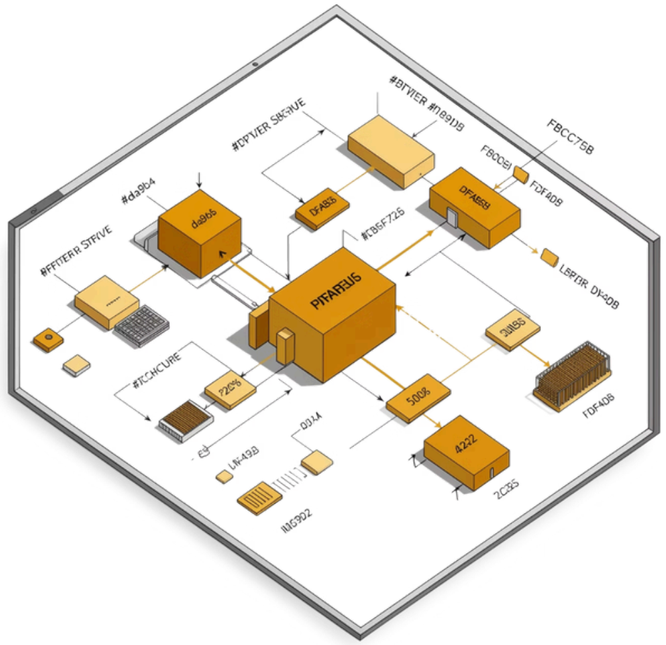




DEMO TIME!



Tech Stack



Flutter + Node.js + PostgreSQL

Cross-platform UI, scalable backend, type-safe schema via Prisma ORM

Firebase

Authentication & real-time push notifications

Clean Architecture

Controller → Service → Repository across 14 backend services



Future Plans

Intelligence & Automation

AI Itinerary Planning

OpenAI API generates optimized schedules from budget & preferences

OCR Receipt Scanning

Auto-extract totals & line items from physical receipts

Multi-Currency Support

Log in local currencies, auto-convert to home currency



Future Plans

Scalability & Experience

Offline Persistence

Local caching & background sync - works in remote areas

Unified Web Dashboard

Full-featured web app for desktop planning & power users

Interactive Map Integration

Plot destinations & track member locations in real time





Key Lessons Learnt

1

Team Spirit

Managing a project with a team of 10 people for the first time made us realize the importance of teamwork and accepting and respecting others' point of views.

2

Handling Real-World Edge Cases

Splitting expenses fairly is more complex than expected

3

Security in Payment Systems

Handling transactions taught the importance of secure APIs and data protection

4

Need for Reliable Data Sync

Keeping expenses and payments consistent across users requires careful state management

THANK YOU!

Any Questions?

